

CSE 411: COMPUTER GRAPHICS

REPORT

Project: Project #2 - Utah Teapot

1. Project Purpose and Summary

The purpose of this project is to develop an interactive simulation of the "Utah Teapot," one of the classic models of computer graphics, using OpenGL and GLUT libraries. Within the scope of the project, it is aimed to comprehend geometric transformations, matrix management, and user interaction. In accordance with the requirements, two different windows running simultaneously were created, and to add extra realism, bonus features such as lighting, surface materials, and depth buffering were integrated into the project.

2. Project Requirements and Implementation

The basic features of the application developed in accordance with the project guidelines are detailed below:

- **First Window (Free Rotation):**

- The first window of the project is programmed to allow the teapot to rotate around its own axis when the left, right, or middle mouse buttons are pressed and moved.
- "Perspective View" is used as the appearance, and the sense of spatial depth is successfully reflected.

- **Second Window (Fixed Angles and Menu):**

- In this window, the left mouse button is disabled for interaction; instead, a pop-up menu attached to the right mouse button is designed.
- When the user right-clicks, options are provided to view the teapot from the Top, Bottom, Left, and Right.
- In order to distinguish the views more clearly from each other, a different color assignment is made for each angle (Top: Green, Bottom: Yellow, Left: Blue, Right: Turquoise).

- **Bonus Features (Layer of Realism):**

- In reference to the "Bonus" section in the project guidelines, the teapots are drawn using `glutSolidTeapot` (solid model) instead of `glutWireTeapot`, and light sources and surface materials are applied to them.
- To achieve realistic porcelain/metallic reflections, `GL_LIGHT0` and `GL_SHININESS` properties are carefully configured.
- `GL_DEPTH_TEST` (depth buffering) is enabled to prevent overlapping surface errors in the image.

3. Code Structure and Function Descriptions

A modular and understandable structure is used in the source code of the application, and the tasks of the main functions are summarized below:

- **initLighting():** This is the function that constitutes the bonus part of the project. It initiates the depth test, places the main light source (GL_LIGHT0) into the scene, and defines the ambient, diffuse, and specular properties of the light. It contains the necessary surface material settings (glMaterialfv) for the teapot to shine like porcelain.
- **reshape(int w, int h):** It balances the 3D camera angle and the viewport using the gluPerspective function when the windows are resized or initially opened.
- **display1() & display2():** These are the rendering loops of the two different windows. display1 applies glRotatef matrix transformations using the angleX and angleY variables dependent on mouse movements. display2, on the other hand, determines the camera position with gluLookAt and the teapot color with glColor3f based on the value selected from the menu using a switch-case structure.
- **mouse1() & motion1():** These are the mouse listeners assigned only to the first window. While mouse1 captures the moment the button is pressed, motion1 dynamically updates the rotation angles of the object by calculating the pixel difference passed during dragging.
- **menu(int value):** It is the controller of the menu that opens when right-clicked in the second window. It assigns the selected value to a global variable and refreshes the screen with glutPostRedisplay().
- **main():** It is the main function block that determines the window sizes, their initial positions on the screen, and starts the GLUT loop (glutMainLoop). Flickering in animations is prevented by using double buffering (GLUT_DOUBLE).

4. Experiences and Challenges Encountered

During the development process of this project, grasping the state machine logic of OpenGL was one of the most important achievements. The main challenges I faced and my solutions are as follows:

1. **Multiple Window Management:** Ensuring that two windows run smoothly within the same loop was difficult at first. To solve this, I explicitly specified to OpenGL which window I was operating on by using the glutSetWindow(windowID) command before each render operation (glutPostRedisplay()).
2. **Camera and Matrix Control:** Teleporting the camera to specific angles around the teapot in the second window (with the gluLookAt function) allowed me to deeply understand how matrices are reset (glLoadIdentity) and reloaded.
3. **Lighting and Material Settings:** Initially, when light was added, the teapots appeared in a flat and "blown out" whiteness. I managed to achieve that sharp porcelain shine and dark, realistic shadows by lowering the ambient brightness of the light source and increasing the reflection coefficient (GL_SHININESS) of the material to a high value like 100. Additionally, I used the glEnable(GL_NORMALIZE) function to prevent the distortion of the light's angles of impact on the surface while the teapot rotates.

5. References

In order to avoid plagiarism and make correct implementations during the project process, the following resources were utilized:

1. CSE 411 Computer Graphics Projects Document (Project Guidelines and Expectations).
2. OpenGL Architecture Review Board, Dave Shreiner (ed.), *OpenGL Reference Manual: The Official Reference Document to OpenGL* (Blue Book), Addison-Wesley.
3. Mason Woo, Jackie Neider, Tom Davis, Dave Shreiner, *OpenGL Programming Guide: The Official Guide to Learning OpenGL* (Red Book), Addison-Wesley.